

ARGUS-1

AI-Enabled Space Domain Awareness: In-Orbit Demonstration on a 6U CubeSat

Submitted for the Space Faculty CubeSat programme. This is my mission design answering the SpaceYZ AI-camera IOD brief. The payload I am flying is the camera from the brief: 1U, 5 V, 10 W, 1.3 kg.

(1) As space hardware is mostly priced “by quote,” so where a vendor has not published a public number, I have said *quote required* and given the product link.

(2) I started playing Kerbal Space Program about a week ago, so if my orbital insertion or bad launch management in the ksp simulation makes a real flight-dynamics engineer cry, that is why. The actual math is in Appendix A so you can check my working.

- Name - Mihir Sunil Shah
- Gmail - mihirsunilshah@gmail.com
- LinkedIn - [linkedin.com/in/mihirsunilshah](https://www.linkedin.com/in/mihirsunilshah)
- Currently a Year 3 Computer Science undergraduate with a minor in Astronomy and wish to help out for the 3U cubesat mission as my FYP.

Contents

1. Executive Summary	2
2. Mission Objectives	2
3. CubeSat Platform Trade Study	3
4. Subsystem Architecture	3
5. Orbit Design	5
6. Launch Strategy	5
7. Team and Personnel	6
8. Project Management, Schedule and Gantt	6
9. Assembly, Integration and Testing	9
10. Ground Operations	9
11. Cost Plan	9
12. Risk Register (Top 10)	10
13. Regulatory and Standards Compliance	10
14. Caveats and Open Questions	11
Appendix A: Orbit and Power Calculations	12
Appendix B: Costing Basis	13
Appendix C: Kerbal Space Program Launch Rehearsal	14
Appendix D: ExoAtlas Orbit Visualisation	16

1. Executive Summary

ARGUS-1 is an 18-month In-Orbit Demonstration (IOD) of SpaceYZ's AI imaging payload, aimed at the Space Domain Awareness (SDA) market. From a 550 km sun-synchronous orbit it detects, tracks and classifies non-cooperative Resident Space Objects (RSOs), running the machine-learning inference **on board** so that only the useful detections are sent down. The point is to give investors hard evidence that the camera is commercially viable for SDA constellation customers such as LeoLabs, Slingshot Aerospace etc.

Headline configuration

- **Platform:** 6U bus with a deployable solar array (about 28 W orbit-average power).
- **Compute:** two tiers, a small always-on bus computer (GomSpace NanoMind A3200) plus a KP Labs Leopard 3-TOPS AI processor that only wakes up for imaging.
- **Pointing:** Blue Canyon XACT-15 attitude control for sub-0.01° accuracy.
- **Comms:** UHF for commands/telemetry, X-band downlink at 50+ Mbps.
- **Launch:** SpaceX Transporter rideshare to SSO, base ticket [S\\$448,000 \(spacex.com/rideshare\)](https://www.spacex.com/rideshare), in an ISISPACE QuadPack deployer.
- **Ground:** KSAT Lite as the operational primary, AWS Ground Station as paid backup, SatNOGS as a near-free redundancy layer.

Why it matters

The investor case rests on three things this mission has to prove:

- (1) the AI camera survives and works in low Earth orbit,
- (2) doing inference on board cuts the downlink volume by 50-100× versus sending raw images, which is the economic differentiator for any SDA constellation, and
- (3) a small team can deliver a flight-ready spacecraft on a Series-A budget. ARGUS-1 is built to retire all three on one flight.

On cost: The honest result is that a fixed program total cannot be quoted from public prices, because the expensive items (AI processor, ADCS, OBC, radios, deployable array, ground passes) are all **quote-only** in this industry. Section 11 shows exactly what I could source to a public, linked price in SGD, and a clear RFQ register for everything else.

2. Mission Objectives

Three nested objectives, weighted as the brief asks. O1 is the must-have, O2 still produces valuable IOD data if O1 underperforms, and O3 is a stretch goal that lifts the return if everything works.

ID	Objective	Success criteria	Weight
O1 (Primary)	On-board AI detection and tracking of RSOs	Detect ≥ 100 unique RSOs over 6 months at $\geq 80\%$ precision. Produce processed detections at ≥ 1 Hz. Full end-to-end downlink of metadata plus image chips.	100%
O2 (Secondary)	Validate the inference-vs-downlink economics	Achieve $\geq 50\times$ reduction in downlinked data versus raw imagery while keeping catalog accuracy. Characterise the false-alarm rate against solar phase angle.	+10%
O3 (Tertiary)	Bonus characterisation and catalog contribution	Contribute observations to public catalogs (USSF 18 SDS or LeoLabs). Characterise sensor performance. Update the ML model in orbit over the air.	+10%

Mission success = 100% of O1. With O2 and O3 the IOD reaches **120%**, which is the threshold for the SpaceYZ Series-B story.

2.1 Heritage missions I am copying from

- **Aerospace Corp. Slingshot 1** (12U, Blue Canyon bus, 2022), which proved the fast modular-payload IOD pattern.
- **NorthStar “Skylark”** (4 satellites by Spire, 2024, 530 km), the first commercial space-based SSA service.
- **KP Labs Intuition-1** (2023) and **ESA Øsat-2** (2024), flight heritage for the Leopard AI processor I picked.

3. CubeSat Platform Trade Study

3.1 Why 6U

Think of the form factor as a budget for three scarce things: power, pointing precision and volume. A 3U is too tight on all three once the 1U camera is in. A 12U is over-specified and eats the money. 6U is the sweet spot.

Criterion	3U	6U (selected)	12U
Payload volume after 1U camera	~1.5 U	~4 U	~10 U
Avg power (deployable panels)	8-12 W	22-30 W	45-60 W
Pointing achievable	0.1-1°	0.003-0.01°	0.002°
Bus mass	~2.0 kg	~4.2-4.8 kg	~11.2 kg
Verdict	Too tight	Best fit	Too costly

The camera draws 10 W, the bus another 12-15 W, and the Leopard processor peaks near 20 W. Sustaining a ~22 W orbit-average load needs the 6U deployable array, and the sub-0.1° pointing needed to hold an RSO in frame needs the XACT-15, which only fits in 6U or larger.

3.2 Candidate buses (the ones I compared)

Bus / vendor	Pointing	Notes / where to look
ISISPACE 6U (priceable parts)	0.1°	Structure is publicly priced, see Section 4.6 BoM
NanoAvionics M6P	~0.1°	nanoavionics.com (quote only)
GomSpace 6U	0.5-1°	gomspace.com (quote only)
Blue Canyon X-SAT Saturn	<0.01°	bluecanyontech.com, best pointing but too costly

Recommendation: build the bus around the ISISPACE 6U structure (it has a public price, see Section 4.6) with an upgraded ADCS interface for the XACT-15. The structure is one of the few big items I can actually put a price on.

4. Subsystem Architecture

ARGUS-1 carries the minimum set of subsystems that still closes every requirement, and nothing extra. Command & data handling (the computers) runs the spacecraft and the AI payload. Attitude control (ADCS) points the camera precisely enough to track an RSO. Electrical power (EPS + array) feeds everything through eclipse. Communications carries commands up and detections down. The structure and thermal subsystem holds it together and keeps the electronics in temperature. I deliberately leave out **propulsion**: at 550 km the orbit decays naturally inside the 25-year rule, so a thruster would add cost, mass and a failure mode for no benefit.

4.1 On-board computers: the two-tier pattern

Every flight-proven AI CubeSat (Intuition-1, Φsat-2, HYTI) uses the same trick: a small, always-on bus computer for the safety-critical housekeeping, plus a powerful AI processor that is power-gated and only switched on when the payload runs. Splitting them isolates the safety-critical side from the Linux-based AI side and keeps idle power under 2 W.

Bus computer: GomSpace NanoMind A3200. Runs command & data handling, fault detection/recovery and telemetry on FreeRTOS. AVR32 at 64 MHz, 32 MB RAM, ~0.4 W, ~20 krad tolerance, hundreds of flight units. As a CS student the part I care about: it runs **NASA core Flight System (cFS)**, an open-source (Apache-2.0) framework with heritage on LRO, MMS and Orion. cFS gives ready-made apps for file transfer, housekeeping and scheduling, and I write four custom apps (telecommand, payload manager, ADCS bridge, FDIR).

OBC product page: gomspace.com/product/nanomind-a3200 + Flight software: [github.com/nasa/cFS \(free\)](https://github.com/nasa/cFS)

Payload AI processor: KP Labs Leopard. AMD/Xilinx Zynq UltraScale+ MPSoC (4× Cortex-A53 + FPGA fabric), up to 3 TOPS at INT8, 16 GiB ECC DDR4, redundant SSD, 5-20 W depending on workload, 64-bit Linux with the Vitis AI toolchain, under 1U. Flight heritage on Intuition-1, Φsat-2 and IGNIS. I chose it because 3 TOPS gives roughly 10× headroom for YOLOv5n/v8n inference at 1 Hz, the FPGA fabric does image preprocessing in hardware, and models can be updated in orbit. Cheaper boards (KP Labs Antelope, Xiphos Q8) stay as fallbacks.

Payload product page: kplabs.space/leopard (no public price, quote required)

4.2 The image-processing pipeline

Capture (camera, raw frames at 1 Hz) → preprocess on the Leopard FPGA (hot-pixel mask, debayer, star mask) → INT8 YOLO inference producing boxes + confidence → multi-frame Kalman tracking → the bus computer picks which image chips are worth downlinking → X-band. The result is about **5 MB of curated data per orbit versus ~500 MB of raw imagery, a 100× reduction.**

4.3 Attitude control: Blue Canyon XACT-15

- Integrated star tracker, three reaction wheels, magnetorquers, sun sensor and IMU in 0.5U. About 0.85 kg, ~3 W. (the star tracker is the more expensive piece of equipment as mentioned in the workshop and 3 reaction wheels is exactly what's necessary for movement on all axes)
- Pointing of 0.003-0.012° demonstrated on orbit (MinXSS-1).
- Cheaper fallback: CubeSpace CubeADCS Gen2 (~0.1°). Vendor: bluecanyontech.com (quote required)

4.4 Power: EPS + 6U deployable array

- Triple-junction cells, body-mounted plus a deployable wing. About 28 W orbit-average, ~60 W peak.
- 8-cell Li-ion pack, 77 Wh, sized for the worst-case ~35-minute eclipse with 30% margin.

The orbit-average power budget closes with about 40% positive margin. The full line-by-line budget is in **Appendix A** so it does not clutter this page. The short version is that I need ~19.6 W and the array provides ~28 W.

4.5 Communications

- **TT&C (UHF):** 400-438 MHz, ≤19.6 kbps, AES-256, commands up, beacon/telemetry down.
- **Payload downlink (X-band):** 7.9-8.4 GHz, 150 Mbps capable, DVB-S2X/CCSDS.
- The X-band link budget (KSAT 3.7 m dish, 10° elevation) closes at ~50 Mbps with QPSK, meeting the 1 GB/day target in 3-4 passes. Link-budget figures are in Appendix A.

4.6 Hardware Bill of Materials

Subsystem	Vendor / product	Price (SGD)	Source link
6U structure	ISISPACE 6-Unit structure	S\$11,846	cubesatshop.com -> 6-unit-cubesat-structure
Bus computer	GomSpace NanoMind A3200	Quote only	gomspace.com -> nanomind-a3200
AI processor	KP Labs Leopard	Quote only	kplabs.space -> leopard
ADCS	Blue Canyon XACT-15	Quote only	bluecanyontech.com
EPS + battery	ISISPACE / GomSpace EPS	Quote only	cubesatshop.com -> electrical-power-system
6U deployable array	EXA DMSA / ISISPACE	Quote only (€4400 - €20000)	cubesatshop.com -> solar-panels
UHF + X-band radios	ISISPACE / EnduroSat	Quote only	cubesatshop.com -> communication-systems
VHF/UHF antenna	ISISPACE 6U/12U deployable	Quote only	cubesatshop.com -> 6u-12u antenna

Firm, sourced hardware so far: S\$11,846 (the structure). Everything else needs an RFQ, see Section 11 for how I would handle the total.

5. Orbit Design

Selected orbit: 550 km circular, sun-synchronous, local time of ascending node 10:30, inclination ~97.6°. Operational life 18 months. It decays naturally in ~6-8 years, satisfying the 25-year disposal rule with no deorbit hardware.

Why this orbit: a sun-synchronous orbit gives the AI detector consistent lighting on every pass, which keeps detection performance stable. 550 km is high enough for an 18-month mission but low enough to decay safely without a thruster, and the 10:30 local time keeps the Sun out of the camera's field while still lighting the targets.

Headline numbers (full derivations in **Appendix A**): orbital period **≈95.6 min**, velocity **7.59 km/s**, **15.06 orbits/day**, sun-sync inclination **97.6°**, max eclipse **~35.6 min/orbit**, ground-track shift **~23.9° west per orbit**.

6. Launch Strategy

6.1 Primary: SpaceX Transporter rideshare

- **Price (public, 2026):** [S\\$448,000 for up to 50 kg to SSO, then S\\$8,960/kg beyond \(spacex.com/rideshare\)](https://spacex.com/rideshare). ARGUS-1 at ~6 kg sits well under the base ticket.
- **Target:** Transporter-19 or -20 (mid-to-late 2027, after the 15-18 month build).
- **Deployer:** ISISPACE 6U QuadPack, release velocity <1.8 m/s. [isispace.nl -> quadpack-deployer](https://isispace.nl/quadpack-deployer) (quote required).

Why rideshare, not a dedicated launcher: a dedicated small launcher costs several times the rideshare ticket for a 6 kg payload. Transporter is a proven, low-cost route to exactly the SSO I need. Note that Singapore has no orbital launch site, so a rideshare flown from the US plus standard export paperwork is the realistic path from here.

6.2 Backups and contingency

Provider	Link	Notes
Rocket Lab Electron (assuming space faculty has some connects as there was a representative at the ATxSG event)	rocketlabusa.com	Dedicated SSO, higher \$/kg
ISRO PSLV rideshare	nsil.gov.in	Cost-competitive, longer lead
Exolaunch (broker)	exolaunch.com	Transporter aggregator, storage if slipped

Rideshare slips of 3-6 months are normal, so the schedule holds a 4-month float. Launch-vehicle insurance is skipped given Falcon 9 reliability. Third-party liability + short in-orbit cover is bought by broker quote (Section 11 marks it RFQ).

7. Team and Personnel

Sized for an 18-month 6U IOD: sixteen positions blended to about 10 full-time equivalents (FTE). It is deliberately software-heavy because the mission's value lives in the AI inference pipeline, and five of the sixteen roles are software.

Discipline	Headcount	Blended FTE	Where the weight sits
Mechanical / Thermal / AIT	3	1.8	Structures, thermal, integration
Computer Science / Software	5	3.6	Bus SW, AI/ML, FPGA, ground, DevOps
Electrical / RF / ADCS	3	1.7	Power, comms, attitude control
Systems & Management	3	1.9	PM, systems engineering, ops
Quality & Test (part-time)	2	0.9	QA, environmental test
Total	16	~9.9	Matches the budget envelope

With an off-the-shelf bus and ADCS, the mechanical and electrical work is mostly integration rather than design, so those teams stay lean. The differentiator is the on-board AI, so software gets the most headcount. Two roles (ADCS, QA) are part-time consultants to keep burn rate down.

7.1 Roles in brief

- **M1 Lead Mechanical/Structures (1.0):** mass budget, deployable stowage.
- **M2 Thermal (0.5):** hot/cold model, radiator + heater sizing, TVAC plan (the 20 W processor in 1U is the challenge).
- **M3 AIT Technician (1.0, Phase D):** physical assembly, harness routing, torque records.
- **S1 Flight Software Lead, bus computer (1.0):** owns the cFS safety-critical core (FDIR, modes, telemetry, command auth).
- **S2 AI/ML Engineer (1.0):** model fits the Leopard, runs at 1 Hz under 50 ms, ≥80% precision, OTA-updatable.
- **S3 Payload Firmware/FPGA (0.8):** Yocto Linux image, Vitis AI runtime, camera drivers, FPGA preprocessing.
- **S4 Ground Software (0.6):** Yamcs mission control, telemetry definitions, dashboards, KSAT/AWS integration.
- **S5 DevOps/Simulation (0.4):** CI/CD, FlatSat orchestration, software-in-the-loop simulator, signed flight images.
- **E1 Lead Electrical/EPS (0.8):** power-budget owner and EPS commissioning (EPS is the most common cause of CubeSat death).
- **E2 RF/Comms (0.5):** UHF + X-band, link budgets, and the ITU/FCC filing (a 9-12 month critical-path item).
- **E3 ADCS (consultant, 0.3):** custom detumble/sun-point/target-track modes and Kalman tuning.
- **P1 Project Manager (0.6):** schedule, budget, risk, procurement, investor comms.
- **P2 Lead Systems Engineer (1.0):** the only person who sees the whole spacecraft: requirements, ICDs, V&V, review chair.
- **P3 Mission Operations Lead (0.5):** contact scheduling, anomaly playbook, the 30-day LEOP.
- **Q1 Product Assurance (0.25) and Q2 Test/AIT Engineer (0.4, Phase D):** non-conformance tracking and the environmental-test evidence package.

8. Project Management, Schedule and Gantt

8.1 Lifecycle phases

Phase	Months	Activities	Exit review
O/A	M1-M2	Requirements, trade studies, concept architecture	Mission Concept Review
B	M3-M5	Subsystem designs, ICDs, simulator stand-up	PDR (M5)
C	M6-M8	Detailed design, long-lead procurement, EM bring-up	CDR (M8)
D	M9-M14	EM/FM build, environmental test campaign	FRR (M14)
Launch prep	M14-M16	Ship, rideshare integration, LV verification	Pre-launch authorisation
E	M16-M18+	LEOP, commissioning, science ops	Closeout report

8.2 Critical path

- **ITU/FCC frequency coordination:** 9-12 months, started in M2, the first thing on the path.
- **Long-lead procurement** (Leopard and XACT-15, ~16 weeks each), ordered at PDR (M5).
- **AI model training and Vitis AI quantization** runs parallel to hardware from M3.

The full month-by-month Gantt is on the next page.

8.3 Master schedule (Gantt)

Light grey = scheduled work , dark grey = critical path , (+) = milestone review. Months run left to right, M1-M18.

Task / phase	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Phase O/A: concept & SRR																		
Requirements baseline & MCR		+																
ITU/FCC frequency coordination																		
Phase B: preliminary design					+													
Long-lead procurement (Leopard, XACT-15)																		
Phase C: detailed design								+										
AI model training (YOLO + synthetic)																		
Vitis AI INT8 quantization + Leopard port																		
Phase D: EM build & cFS bring-up																		
Flight Model (FM) assembly																		
Environmental test (vibe, TVAC, EMC)																		
Flight Readiness Review														+				
Ship to launch site & integration																		
Launch (SpaceX Transporter)																+		
LEOP (30 days) + commissioning																		
Phase E: science ops + final report																		

8.4 Milestones and payment schedule

Milestone	Month	Payment	Primary deliverables
Kick-off	M0	15%	Contract signature
Preliminary Design Review	M5	20%	Subsystem PDR packs, requirements baseline, ICDs v1, risk register
Critical Design Review	M8	20%	Detailed design dossier, frozen parts list, V&V matrix
Flight Readiness Review	M14	20%	Environmental test reports, launch package
Launch	M16	15%	On-orbit, first contact
LEOP complete	M17	5%	All subsystems commissioned, model uploaded
Closeout	M18+	5%	Performance-vs-objectives report, IOD evidence package

This weighting front-loads cash through CDR (when vendor purchase orders are largest) and reserves 25% for after launch, so the contractor is paid for mission performance, not just delivery.

9. Assembly, Integration and Testing

Testing follows the ECSS and NASA GEVS standards: full-functional and day-in-the-life runs, 8 thermal-vacuum cycles (−20 to +60°C), random vibration (protoflight 14.1 Grms across three axes), shock per the Falcon 9 user guide, an EMI/EMC suite, and deployment tests of the array and antennas.

On AIT costs: cleanroom, thermal-vacuum, vibration and EMC facility hire are all quoted by the facility, so I have not included day-rates. In Singapore these would be sourced from a local lab (I think NUS has a few, and am not sure if Space Faculty has their own clean rooms to be set up) by quote. The one piece of test ground-support-equipment I can price is a 6U mass simulator for fit-checks and as a launch surrogate:

AIT item	Price (SGD)	Source / status
LoadPath 6U mass simulator (fit-check GSE)	S\$8,064	loadpath-6u-mass-simulator">cubesatshop.com -> loadpath-6u-mass-simulator
Cleanroom hire (ISO 8)	Quote only	Singapore lab / university (RFQ)
Thermal-vacuum, vibration, EMC	Quote only	Test house day-rate (RFQ)

10. Ground Operations

- **Primary, KSAT Lite:** 26 sites, 280+ antennas. Per-pass pricing is by quote (not published). [ksat.no -> ksatlite](http://ksat.no->ksatlite) (RFQ).
- **Secondary, AWS Ground Station (publicly priced):** narrowband **S\$3.84/min** reserved (S\$12.80/min on-demand). Wideband S\$12.80/min reserved. AWS has an Asia-Pacific (Singapore) site, which is convenient from here. aws.amazon.com/ground-station/pricing.
- **Tertiary, SatNOGS:** free open network for redundant UHF beacon checks. A basic DIY station is roughly **S\$400** in parts. wiki.satnogs.org/Build.

Mission control software (all free, open source): Yamcs for telemetry/telecommand and OpenC3 COSMOS for the operator UI.

Why three networks: KSAT alone closes every downlink requirement. AWS adds surge capacity with no fixed dish (and a Singapore site). SatNOGS is effectively free insurance for the beacon. Ground-ops usage arithmetic (passes per day, minutes, the AWS-rate computation) is in Appendix B.

11. Cost Plan

The brief wanted a tidy total. The honest answer is that **a fixed program total cannot be quoted from public prices**, because the big-ticket items (AI processor, ADCS, OBC, radios, deployable array, launch deployer, ground passes, test facilities, personnel) are all quote-only in this industry. So instead of a random precise total, here is exactly what I could source to a public link plus the RFQ register for everything else.

11.1 Firm, publicly-sourced line items

Item	Price (SGD)	Source link
ISISPACE 6U structure	S\$11,846	6-unit-structure">cubesatshop.com -> 6-unit-structure
SpaceX rideshare (base, ≤50 kg)	S\$448,000	spacex.com/rideshare
6U mass simulator (AIT GSE)	S\$8,064	loadpath-6u">cubesatshop.com -> loadpath-6u
SatNOGS DIY redundancy station	~S\$400	wiki.satnogs.org/Build
Flight + ground software (cFS, Yamcs, COSMOS)	S\$0	github.com/nasa/cFS
AWS Ground Station (backup)	S\$3.84/min	aws.amazon.com/ground-station/pricing
Firm one-off subtotal	S\$468,310	structure + launch + sim + SatNOGS + software

11.2 Quote-only register (RFQ at kick-off)

These each need a request-for-quote, and I would issue all of them in Phase 0/A and lock prices at the purchase order. None are guessed here. Links are in Section 4.6, Section 6 and Section 10: KP Labs Leopard, GomSpace NanoMind A3200, Blue Canyon XACT-15, EPS + battery, 6U deployable array, UHF + X-band radios, deployable antenna, ISISPACE QuadPack, KSAT Lite passes, cleanroom + TVAC + vibe + EMC facility hire, personnel, and insurance.

Recommendation: carry the firm sourced items as fixed, issue the RFQ register on day one, and report a single consolidated total to investors only once real quotes are in. That is slower than quoting a round number now, but it is the version I can defend line by line.

12. Risk Register (Top 10)

#	Risk	Likelihood	Impact	Mitigation
1	Rideshare launch slip of 6+ months	H	M	Multi-mission booking via Exolaunch, storage plan
2	AI model sim-to-real gap on orbit	M	H	Synthetic data + STK-EOIR, over-the-air model updates
3	Leopard supply lead time (16+ wk)	M	H	Order at PDR, Xiphos Q8 fallback evaluated
4	Pointing performance shortfall	M	H	XACT-15 (top tier), on-orbit calibration
5	Camera-Leopard interface delays	M	M	Early EM-to-EM integration test at M9
6	ITAR/EAR controls on XACT-15	L	M	Jurisdiction review at PDR, export paperwork to Singapore early
7	ITU/FCC coordination delays	H	M	File at M2, first item on the critical path
8	Battery degradation in eclipse cycling	L	M	30% EPS margin, cell-balancing firmware
9	Radiation dose exceeds margin at EOL	L	L	All electronics ≥12 krad, mission ~5 krad
10	Series-A burn rate vs schedule	M	H	Milestone-gated payments, pursue co-funding

13. Regulatory and Standards Compliance

- **CubeSat Design Specification Rev 14.1** (Cal Poly), structural and interface compliance. cubesat.org/cubesatinfo
- **ECSS-E-ST-10-03C Rev.1**: environmental test standard. **NASA GEVS** for random vibration and TVAC.
- **IADC 25-year disposal rule**: met by natural decay from 550 km, no deorbit hardware.
- **ITU frequency coordination**: UHF amateur via IARU, commercial X-band via the FCC (9-12 months, starts at PDR).
- **ITAR / EAR**: XACT-15 (US) and Leopard (EU) need a dual-use jurisdiction review. As the buyer is in Singapore, export licensing is started early.

14. Caveats and Open Questions

- KP Labs Leopard mass and radiation rating are not public, and the vendor must confirm them in writing at PDR.
- Almost all flight hardware is quote-only, so the program total in Section 11 is deliberately left open until RFQs return.
- Vendor inflation has run 8-12%/year and lead times have stretched to 9-14 months, so all prices lock at the purchase order.
- AI detection performance depends heavily on synthetic-training-data fidelity, so expect 30-50% recall on the first orbit, improving via on-orbit updates.
- All hardware is imported into Singapore, so freight and ~9% GST sit on top of the ex-works prices above (noted in Appendix B).
- We would probably need multiple of some components for redundancy + testing, have not included here as I tried to get the budget to a minimum.

Appendix A: Orbit and Power Calculations

All the maths the body skipped. Constants: $\mu = 398,600 \text{ km}^3/\text{s}^2$, $R_E = 6,378 \text{ km}$, $J_2 = 1.0826 \times 10^{-3}$, semi-major axis $a = R_E + 550 = 6,928 \text{ km}$, eccentricity $e \approx 0$.

A.1 Orbit

Parameter	Formula	Result
Orbital period	$T = 2\pi \sqrt{a^3/\mu}$	5,736 s $\approx$ 95.6 min
Orbital velocity	$v = \sqrt{\mu/a}$	7.59 km/s
Orbits per day	$n = 86,400 / T$	15.06 orbits/day
Sun-sync inclination	$\cos i = -(2 \times \omega_E) / (3 \times J_2 \times (R_E/a)^2 \times n)$	$i = 97.6^\circ$
Max eclipse fraction	$\cos \alpha = \sqrt{1 - (R_E/a)^2}$	37.2% $\rightarrow$ $\sim$ 35.6 min/orbit
Ground-track shift	$360^\circ / 15.06 \text{ orbits}$	23.9° west per orbit
Natural decay (BC $\sim$ 75 kg/m ²)	atmospheric-drag model, nominal F10.7	$\sim$ 6-8 years (meets 25-yr rule)

A.2 Orbit-average power budget (sunlight)

Load	Avg power (W)	Duty
Bus computer + EPS housekeeping	1.5	100%
ADCS (XACT-15)	3.0	100%
UHF beacon (TT&C)	1.5	100%
Leopard processor	5.0	25% $\times$ 20 W
AI camera (1U, 10 W)	2.5	25% $\times$ 10 W
X-band transmit	0.6	5% $\times$ 12 W
Thermal heaters	1.0	duty
Subtotal	15.1	
+ 30% margin = required	19.6	
Provided (array, orbit-avg)	$\sim$28	$\sim$ 40% positive margin

A.3 X-band link (sanity check)

8 GHz, $\sim$ 2 W transmit, 5 dBi patch, free-space path loss $\sim$ 175 dB at 550 km / 10° elevation, ground G/T $\sim$ 33.5 dB/K $\rightarrow$ about 50 Mbps sustained with QPSK rate-1/2. That clears the 1 GB/day requirement in 3-4 X-band passes.

Appendix B: Costing Basis

Everything in the body is presented in Singapore dollars. This appendix shows the one-line currency basis and the ground-ops arithmetic, both of which I kept out of the main text.

- **Currency basis (May 2026):** published vendor prices in USD and EUR were converted to SGD at the prevailing mid-market rate (roughly US\$1 = S\$1.28 and €1 = S\$1.49). Examples: ISISPACE 6U structure €7,950 -> S\$11,846, SpaceX base US\$350,000 -> S\$448,000, mass simulator US\$6,300 -> S\$8,064, AWS narrowband US\$3/min -> S\$3.84/min. Rates move daily, so every figure is re-confirmed at the purchase order.
- **Singapore import:** all flight hardware is imported, so freight plus ~9% GST sits on top of the ex-works prices. AWS Ground Station's Asia-Pacific (Singapore) site reduces ground-segment latency from here.
- **Ground-ops arithmetic (illustrative, on the AWS published rate):** ~3 passes/day × ~8 min × ~540 days ≈ 12,960 antenna-minutes over 18 months. At the AWS narrowband reserved rate of S\$3.84/min that is about S\$49,766, while high-rate X-band passes would bill at the wideband rate. KSAT Lite is the operational primary and is priced separately by quote.

Appendix C: Kerbal Space Program Launch Rehearsal

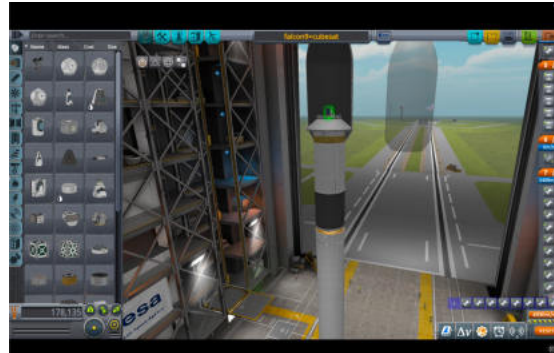
Before I wrote the launch and orbit sections I rebuilt the mission in Kerbal Space Program to get a feel for ascent and orbit insertion. It took me roughly 50 attempts and a lot of evenings, but I eventually flew a Falcon 9 replica carrying a 1U CubeSat all the way to orbit. The full launch video goes in the submission email (I'll try to attach it as its quite large and gmail has a limit). The screenshots below walk through what actually happened.

C.1 Building the vehicle

I assembled the Falcon 9 and its payload in the Vehicle Assembly Building (VAB). The booster, second stage, fairing and CubeSat are all stacked as one launch vehicle.



The assembled Falcon 9 plus payload in the VAB.



The 1U CubeSat mounted under the fairing (highlighted green).

One honest gap: I designed a 6U CubeSat separately to match the real mission, but the parts I built were not close enough to scale to attach it cleanly to this booster, so I flew a 1U for the demonstration instead. The 6U build is shown below on its own. If I can somehow figure out how to attach the 6U cubesat to my falcon9, I will resend an updated video if required.



The 6U analog, built separately. The flown mission design stays 6U, the demo flight used a 1U.

C.2 Ascent and staging

Launch was clean. The first stage burned until separation at about T+2 minutes and 60 km altitude. I made one deliberate compromise: I did not keep any fuel back to land the booster. I could not get the landing controls right, so I spent the booster propellant getting to orbit faster rather than attempting a recovery. On the real mission the booster recovery is the launch provider's job, not mine, so this does not change the proposal.



First-stage separation at roughly T+2 min, around 1350 m/s.

C.3 Orbit insertion and deployment

After staging I set up a circularization burn with a maneuver node, flew it, and separated the CubeSat. The final orbit came out near 747 km apoapsis by 406 km periapsis, period about 1h 11m. I aimed for a 550 km circular sun-synchronous orbit and missed: mine is higher and more elliptical than intended. A clean circular SSO is the main thing I would practice next.



Planning the circularization burn (a 1120 m/s node).



Second-stage and CubeSat separation at about T+13 min.



Deployed CubeSat with its antenna live. Final orbit about 747 km by 406 km, period about 1h 11m.

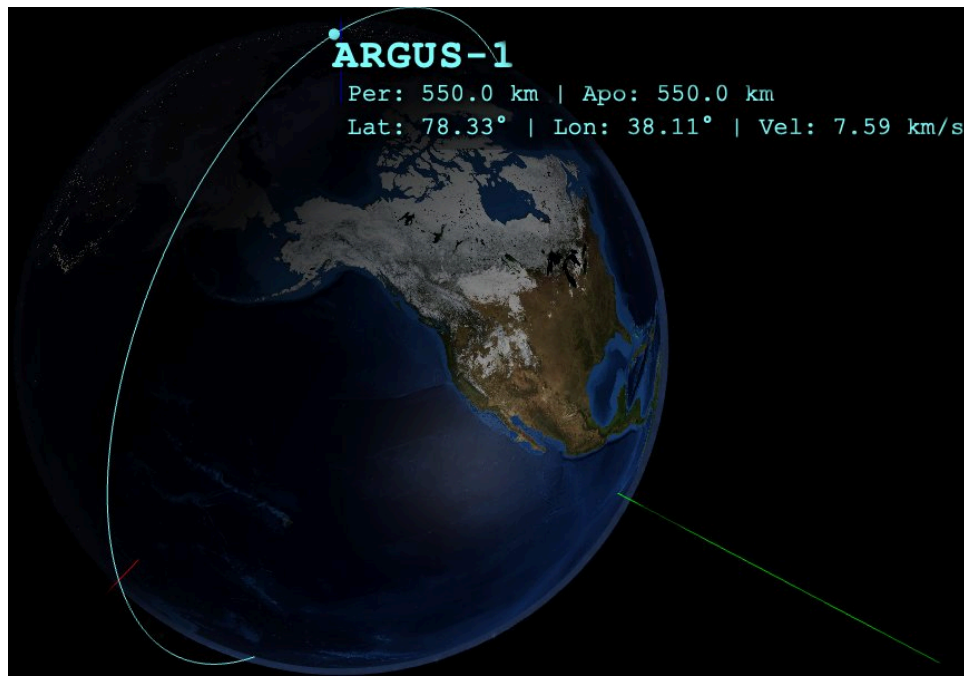
C.4 What I learned and would fix

- **Orbit accuracy:** I reached orbit but not the 550 km circular SSO I wanted (I ended near 747 km by 406 km). Next time I would plan the insertion more carefully and circularize properly at apoapsis.
- **Booster recovery:** I flew the first stage as expendable because I could not land it. Worth practicing, even though the real launch provider handles recovery.
- **Payload scale:** I could not attach a 6U to my booster cleanly, so I demonstrated with a 1U. The flown mission design stays 6U.
- **Tooling:** KSP is great for intuition but not for real numbers. For the flight design I tried to validate the orbit in a proper tool such as NASA GMAT but the software isn't working well on my mac so I just found few online tools for orbit simulation and made my own 2D visualization.

Appendix D: ExoAtlas Orbit Visualisation

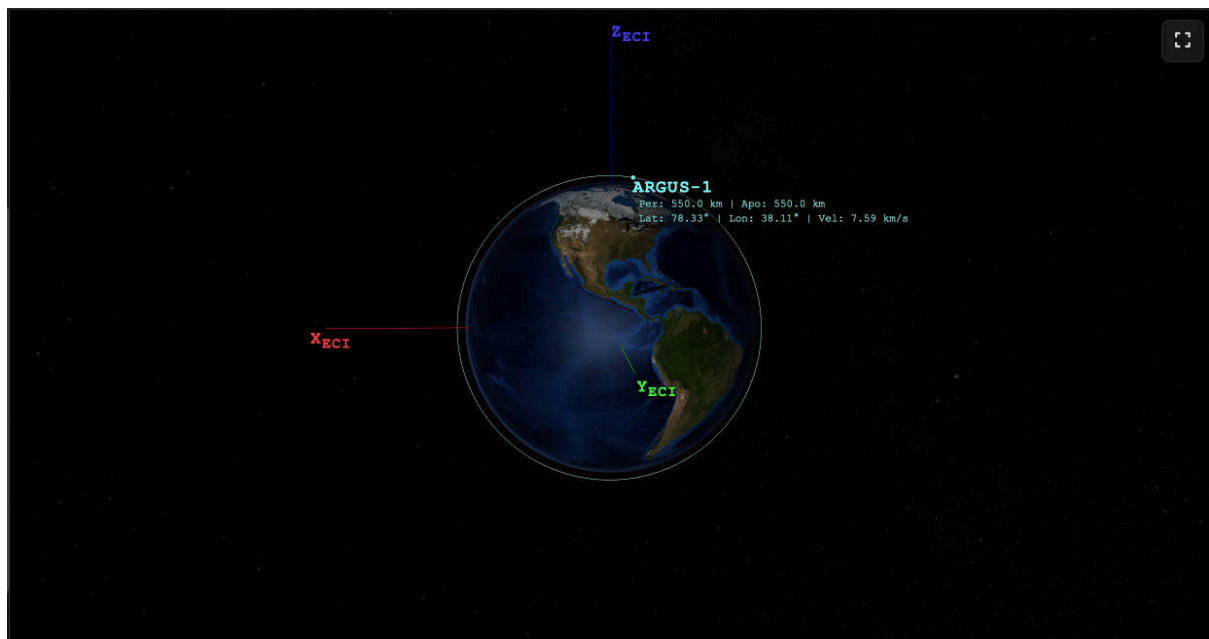
The KSP rehearsal in Appendix C gave useful intuition for the ascent and orbit insertion, but the orbital geometry it produced was approximate. After the KSP work I modelled the intended ARGUS-1 orbit using ExoAtlas Orbit Visualizer, a browser-based SGP4 tool that accepts classical orbital elements as direct input. The figures below show the 550 km, 97.6 degree sun-synchronous orbit used throughout this proposal.

Figure D.1: Near-polar SSO pass over the Arctic



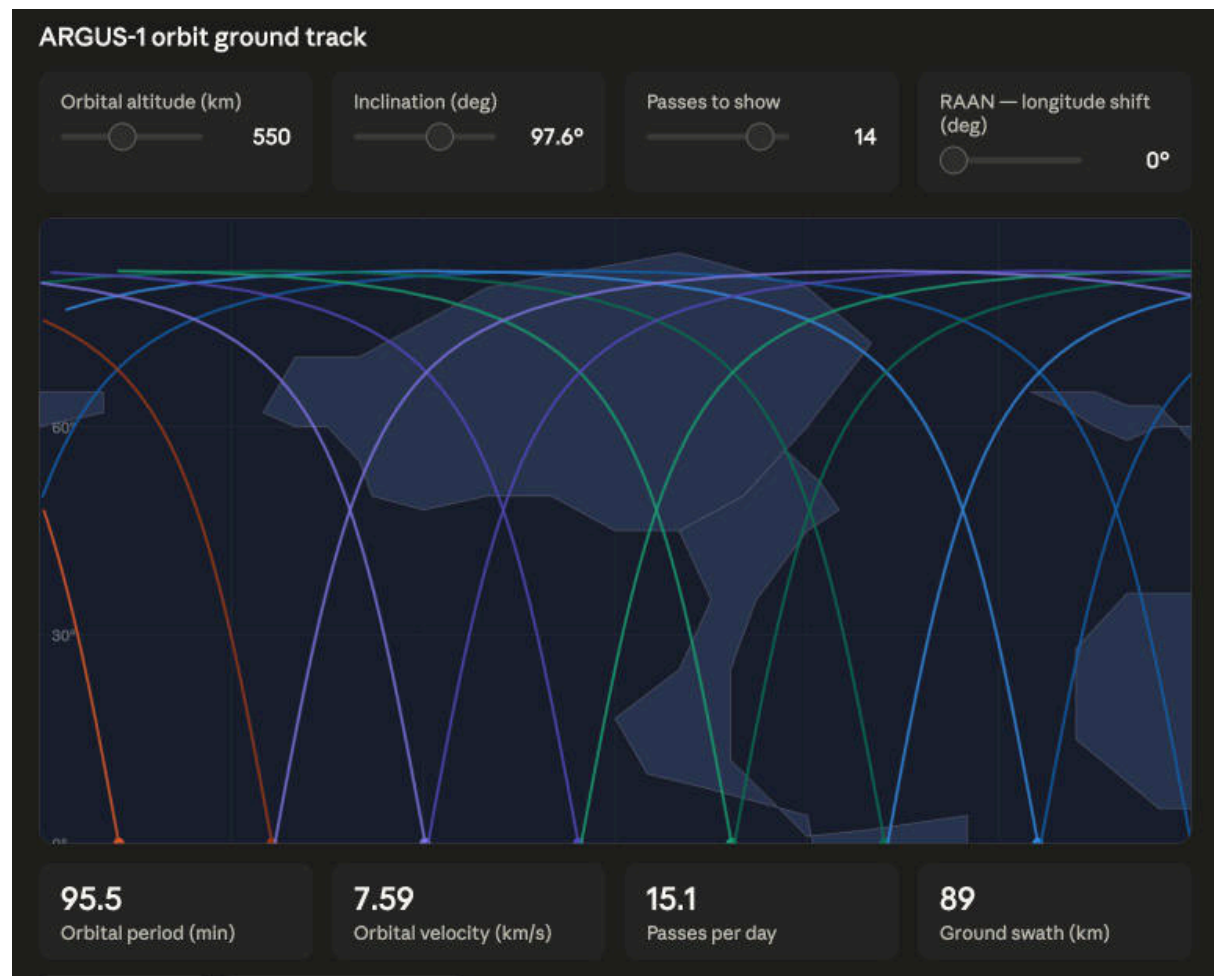
ARGUS-1 at 550 km circular SSO, 97.6 degree inclination, 7.59 km/s orbital velocity. Near-polar pass shown over the Arctic with day/night terminator visible. Generated using ExoAtlas Orbit Visualizer.

Figure D.2: Full orbit in the Earth-Centred Inertial (ECI) reference frame



ARGUS-1 orbit in the ECI reference frame showing the near-polar retrograde geometry. Circular orbit confirmed by equal perigee and apogee at 550.0 km. Generated using ExoAtlas Orbit Visualizer.

Figure D.3: 2D ground track showing global coverage over 14 passes



ARGUS-1 ground track over 14 passes (approximately 22 hours). Orbital period 95.5 min, 15.1 passes per day, ground swath 89 km. The sinusoidal pattern on the Mercator projection is characteristic of a near-polar orbit; successive passes shift westward as Earth rotates beneath the satellite, achieving full global coverage within one day. Generated using Keplerian propagation, 550 km, 97.6 degree inclination.